Long-Horizon Rollout Diagnostics for Latent World Models: LeWM on Push-T

Hanwen Yang
Independent Researcher
Melbourne, VIC, Australia
kevin.yang0047@gmail.com

© 2026 IEEE. Accepted for publication in the 2026 International Conference on Digital Image Computing: Techniques and Applications (DICTA 2026), Sydney, Australia, 2–4 December 2026. Proceedings forthcoming. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works.

***Abstract*—Latent world models are increasingly used for prediction and planning, yet current evaluation focuses primarily on representation quality, decoded visualisations, or downstream planning performance rather than the fidelity of the latent rollouts themselves. We ask a simple question: do long-horizon latent rollouts preserve the correct physical evolution of the environment? To answer this, we introduce a rollout-level diagnostic framework that evaluates physical-state decodability, target alignment, and trajectory geometry directly in latent space. We apply this framework to the released LeWM checkpoint on the Push-T benchmark using frozen physical-state probes, latent trajectory analysis, and a teacher-forced control. We find a consistent failure mode: predicted latents remain well-scaled throughout long-horizon rollout, yet progressively lose alignment with the correct future state. Physical-state decodability deteriorates, rollout trajectories become systematically under-updated and overly straight, and the ratio of predicted to true encoded transition magnitude falls from 0.32 to 0.16. Teacher forcing reveals that substantial under-updating is already present in the local one-step predictor, while open-loop autoregressive feedback further increases long-horizon target error. These findings show that norm stability and visually plausible decoded rollouts are insufficient to assess rollout fidelity, and motivate evaluating latent world models using rollout-level measures of state decodability, target alignment, and trajectory geometry.**

***Index Terms*—world models, latent dynamics, representation learning, embodied intelligence, long-horizon prediction**

I. Introduction

World models enable agents to predict the consequences of their actions and plan by imagining future trajectories before acting in the environment [1]–[3]. Recent advances increasingly perform this prediction in latent representation spaces, where compact representations encode predicted futures without requiring pixel-level reconstruction or explicit physical states. Joint-Embedding Predictive Architectures (JEPAs) predict future representations rather than pixels [4]–[6], while related latent world models have shown promising results in robot planning and decision making [7], [8].

Planning with latent world models ultimately depends on the quality of long-horizon latent rollouts. During inference, the predictor recursively feeds its own outputs back as inputs to imagine future trajectories. While prediction errors are expected to accumulate over time [9], [10], it remains unclear how these rollouts degrade and what information is lost during long-horizon prediction. Unlike pixel-space world models, where degradation can often be observed visually as blur or drifting objects [11], latent rollouts are not directly interpretable. A rollout may remain norm-stable or produce visually plausible decoded predictions while no longer corresponding to the correct future physical state.

Current evaluation provides only partial answers. Existing work typically evaluates latent world models through three complementary perspectives: probing physical state from encoded latents, visualising decoded rollouts, and measuring downstream planning performance [7], [8], [12]. These evaluations establish whether the learned representation contains useful information, whether decoded trajectories appear plausible, and whether the resulting planner or policy succeeds. However, they do not directly evaluate rollout fidelity: whether predicted latent trajectories continue to preserve task-relevant physical information and remain aligned with the correct future as prediction horizons increase. Recent work has begun to address long-horizon degradation by proposing improved training objectives or architectures [13], but quantitative analysis of rollout behaviour itself remains limited.

In this paper, we introduce a rollout-level diagnostic framework for analysing latent world models. As a complement to downstream task performance, the framework directly measures how latent rollouts evolve over long horizons through three diagnostics. Frozen physical-state probes test whether rollout latents retain decodable task-relevant state. Latent-trajectory analysis measures target alignment, transition magnitude, and trajectory geometry. Finally, a teacher-forced control tests whether transition-magnitude under-updating is already present in the local one-step predictor while quantifying the additional target error introduced by autoregressive feedback. Together, these diagnostics distinguish representation plausibility from rollout fidelity and provide complementary views of long-horizon latent dynamics.

We demonstrate the framework through a controlled case study of the released LeWM checkpoint on the Push-T benchmark [7]. Across 1,000 held-out episodes, the analysis reveals a consistent pattern: predicted latents remain well-scaled while progressively losing target alignment, with dete-

riorating physical-state decodability and systematically under-updated, overly straight trajectories. Teacher forcing further shows that substantial transition-magnitude under-updating is already present in the local one-step predictor, while recursive rollout introduces additional long-horizon target error. These findings show that apparently well-behaved rollout latents can nevertheless exhibit degraded long-horizon dynamics.

Our contributions are threefold:

- We introduce a rollout-level diagnostic framework for latent world models that jointly evaluates physical-state decodability, target alignment, trajectory geometry, and teacher-forced prediction, complementing representation probes, decoded visualisations, and downstream planning evaluation.
- We quantitatively characterise a failure pattern in the released LeWM Push-T checkpoint: predicted rollouts remain well-scaled while progressively losing alignment with the correct encoded future, accompanied by deteriorating physical-state decodability and systematically under-updated, over-smoothed trajectories.
- We use a teacher-forced control to distinguish transition-magnitude under-updating already present in the local one-step predictor from the additional target error introduced by recursive open-loop rollout.

II. Related Work

Latent world models. Learning a model of the environment and planning inside it is a long-standing idea [1]–[3]. Joint-embedding predictive architectures brought the idea to representation space [4], [5], and several recent systems build action-conditioned world models there: DINO-WM predicts on frozen pretrained features [8], PLDM trains latent dynamics with multiple interacting objectives [12], and LeWM reduces the recipe to a prediction loss plus one regulariser [7]. Among these, we analyse LeWM because this simpler training objective makes it a clean case study for relating training signal to rollout behaviour. Recent work has also begun to address long-horizon latent prediction from different angles. HWM reduces rollout length through hierarchical planning [14]. V-JEPA 2 connects self-supervised video prediction with action-conditioned robot planning [6]. LiveWorld addresses a related issue by making persistent state explicit for out-of-sight dynamics [15]. These works propose models; this paper analyses one.

Long-horizon prediction error. That model rollouts drift over long horizons is well documented for explicit-state dynamics models: short rollouts are a standard mitigation [9], and the shape of compounding error has been studied empirically [10]. Two things change in the latent setting. The state is no longer observable, so degradation must be measured through instruments; it cannot be read off directly. In regularised JEPA-style latent models such as LeWM, the regulariser also constrains marginal encoder statistics, so marginal statistics alone may not reveal whether the rollout tracks the correct future. Sub-JEPA reports qualitative long-horizon drift in LeWM and responds with a modified regulariser [13]. Our paper adds the missing measurement: a quantitative separation between the one-step deficit and the additional error introduced by open-loop feedback.

Probing and diagnostics. Reading information out of frozen representations with small trained probes is an established tool [16], with known caveats about probe capacity and validity [17]. We use probes in a specific rollout setting: they are calibrated on real encoded latents and then applied to predicted latents horizon by horizon. Recent video-world-model benchmarks also evaluate world models by scoring generated video for perceptual and physical fidelity [18]. Unlike rendered-video benchmarks, we evaluate latent rollouts directly, using state readout, target error, and trajectory geometry.

III. Background: LeWM and Rollout Regimes

LeWM [7] is an end-to-end joint-embedding predictive world model [4] consisting of an encoder and an action-conditioned predictor, optimised jointly from pixels using exactly two loss terms: a next-latent prediction loss and a SIGReg regularisation loss, with no stop-gradients, exponential-moving-average targets, or pre-trained encoders. This simplicity makes it an attractive subject for controlled analysis: the training signal has only two parts, so rollout behaviour can be related to them directly. A ViT-Tiny encoder f_θ maps an observation o_t to the 192-dimensional latent

$$z_t = f_\theta(o_t) \in \mathbb{R}^{192}, \tag{1}$$

and a ViT-S predictor g_ϕ, causally masked over a three-frame latent history, predicts the next latent. Observations are subsampled at frame-skip 5, with the five raw actions between retained frames grouped into an action block. We denote by A_t the action blocks aligned with the three-frame latent context and the next prediction step. Here $z_{t-2:t}$ denotes the three-frame latent context (z_{t-2}, z_{t-1}, z_t). We measure h in raw environment steps from the first retained frame. The three context frames are at $h = 0, 5, 10$, and the predictions are at $h = 15, 20, \ldots, 100$. One predictor step is

$$\hat{z}_{t+1} = g_\phi(z_{t-2:t},\, A_t)\,. \tag{2}$$

Both networks are trained on a single objective;

$$\mathcal{L}(\theta, \phi) \;=\; \mathbb{E}\big[\, \|\hat{z}_{t+1} - z_{t+1}\|_2^2 \,\big] \;+\; \lambda\, \mathrm{SIGReg}(Z), \tag{3}$$

. The first term trains the predictor to match the encoded latent of the next observation, while the SIGReg term regularises the marginal distribution of encoded latents toward an approximately isotropic, non-collapsed distribution [19]. Two properties of this objective matter for what follows. First, the regulariser acts on encoder outputs during training, not on open-loop rollout latents. Thus, if predicted latents remain well-scaled during rollout, this only shows that they remain globally plausible under the learned latent distribution; it does not show that they correspond to the correct future state. Second, the prediction loss supervises only single next-latent steps within short training windows. It does not train the

predictor under the long-horizon open-loop condition, where each prediction becomes part of the context for the next step.

At inference, the predictor runs in two regimes that differ only in which latents fill its context. In the *teacher-forced* regime, every context latent in Eq. (2) is a real encoded latent, as during training; the resulting prediction is denoted $\hat{z}^{\mathrm{tf}}_{t+1}$. In the *open-loop* regime, used for latent imagination and planning, only the initial three-frame context is encoded from observations. Subsequent context latents are the predictor's own previous outputs, and the resulting prediction is denoted $\hat{z}^{\mathrm{ol}}_{t+1}$:

$$\hat{z}^{\mathrm{ol}}_{t+1} = g_\phi\big(\hat{z}^{\mathrm{ol}}_{t-2:t},\, A_t\big)\,, \qquad \hat{z}^{\mathrm{ol}}_{s} = z_s \;\text{ on context steps.} \quad (4)$$

Thus, teacher forcing measures the local one-step prediction under real encoded inputs, whereas open-loop rollout measures the behaviour of the predictor when its own predictions are recursively reused.

The horizon gap is central to our analysis. Training supervises one-step prediction within four-frame windows. Under our indexing, the final context frame is at $h{=}10$, so LeWM's five-step receding-horizon planner ends at $h{=}35$, matching its released rollout visualisations. We evaluate eighteen open-loop steps, ending at $h{=}100$, where errors can accumulate without direct supervision of the recursively generated trajectory.

We study the released Push-T checkpoint. In Push-T [20], a circular end-effector pushes a T-shaped block toward a target pose, and the task-relevant physical state $s_t = \big(p^{\mathrm{agent}}_t,\, p^{\mathrm{block}}_t,\, \alpha^{\mathrm{block}}_t\big)$ comprises agent position, block position, and block angle. These are the quantities LeWM's own evaluation decodes from encoded latents with frozen probes; Section IV uses these probes, together with latent-geometry diagnostics, to measure how Eq. (4) behaves as the rollout horizon increases.

IV. Diagnostic Protocol

The protocol asks three questions about the open-loop regime of Eq. (4): whether predicted latents retain decodable physical state, horizon by horizon; how the geometry of predicted trajectories differs from true encoded trajectories; and whether the observed degradation is caused by open-loop compounding or is already present in the local one-step prediction. We address these questions with three corresponding diagnostics: frozen physical-state probes, latent-geometry metrics, and a teacher-forced control. All diagnostics use the Push-T expert dataset released with LeWM, split by episode index into training, validation, and test partitions using a fixed seed. Physical-state probes are trained on encoded latents from the training partition. Reference statistics for latent-norm and empirical-manifold diagnostics are computed from encoded latents in the training and validation partitions. All metrics reported in Section V are computed on 1,000 held-out test episodes. For each episode, the first three retained frames are encoded as context ($h = 0, 5, 10$), and the predictor is rolled forward for eighteen open-loop steps under the recorded expert action blocks, producing predictions at $h \in \{15, 20, \ldots, 100\}$.

TABLE I

Instrument calibration: probe MSE on held-out *encoded* latents ($h{=}0$) under LeWM's protocol, compared with the reported values. Position errors use normalised Push-T coordinates; angle errors use radians. The probes match the published accuracy ordering at comparable magnitudes before being applied to rollout latents.

	Linear		MLP	
Quantity	Ours	[7]	Ours	[7]
Agent location	0.043	0.052	0.005	0.004
Block location	0.021	0.029	0.001	0.001
Block angle	0.175	0.187	0.027	0.021

The recorded actions are supplied unchanged in all conditions; the protocol varies only the source of the context latents.

A. Physical-state probes

To test whether a rollout latent preserves recoverable information about the physical state s_t, we adopt LeWM's probing protocol [7]. Linear and MLP probes p_ψ are trained to regress each component of s_t from *encoded* latents of real frames [16], and are then frozen. Linear probes are affine maps from the 192-dimensional latent to each state component; MLP probes have one hidden layer of width 256 with ReLU. Applying these same probes to predicted rollout latents tests whether predictions retain the physical-state information that is decodable from real encoded latents. This is the relevant test for latent-space rollout, where future states are represented by predicted latents rather than observed frames.

For a state component $q \in \{p^{\mathrm{agent}}, p^{\mathrm{block}}, \alpha^{\mathrm{block}}\}$, let $s^q_{i,h}$ denote the corresponding ground-truth state in episode i at raw horizon h. We decode each open-loop latent with the frozen probe for that component and compare it to the ground truth:

$$\mathrm{NMSE}_q(h) \;=\; \frac{\sum_i \| p_{\psi,q}(\hat{z}^{\mathrm{ol}}_{i,h}) - s^{\mathrm{q}}_{i,h} \|_2^2}{\sum_i \| \bar{s}^{\mathrm{test}}_q - s^{\mathrm{q}}_{i,h} \|_2^2} \quad (5)$$

Here $p_{\psi,q}$ is the frozen probe for component q, and $\bar{s}^{\mathrm{test}}_q$ is the test-set mean of that component. The numerator is the probe error on predicted rollout latents; the denominator is the error of ignoring the latent and always predicting the test-set mean. Thus, $\mathrm{NMSE}_q(h) = 1$ corresponds to the constant-mean baseline; values below 1 indicate useful decodable information about q, and values above 1 indicate worse-than-baseline recovery under the frozen probe.

We use MLP probes as the primary readout because they reproduce LeWM's stronger encoded-latent probing results, and repeat the same measurements with linear probes as a simpler check. Table I validates the probe readout: on held-out encoded latents, our probes reproduce the accuracies reported by LeWM. Degradation measured later on predicted rollout latents is therefore not explained by poor probe performance on real encoded latents.

B. Latent-trajectory geometry

Probe errors measure recoverable physical state, but they do not show how the predicted trajectory moves through latent space. We therefore use five latent-geometry diagnostics.

First, *target error* is the per-dimension squared distance $\frac{1}{d}\|\hat{z}^{\mathrm{ol}}_{i,h} - z_{i,h}\|_2^2$ between a predicted latent and its true future encoded latent, averaged over test episodes. This tests whether the rollout predicts the correct future latent, not just a well-scaled latent.

Second, *norm* tests whether predicted latents remain on the same scale as true encoded latents. We report predicted latent norms as z-scores relative to 2.1M encoded train/validation latents ($\mu_{\mathrm{ref}} = 13.89$, $\sigma_{\mathrm{ref}} = 0.82$). This detects global norm collapse or explosion.

Third, *nearest-neighbour distance* tests whether predicted latents remain close to the empirical encoded-latent manifold. Let $\mathcal{Z}_{\mathrm{ref}}$ be the encoded train/validation reference set. For any latent x,

$$D_{\mathrm{NN}}(x) = \min_{z \in \mathcal{Z}_{\mathrm{ref}}} \tfrac{1}{d}\|x - z\|_2^2, \quad R_{\mathrm{NN}}(h) = \frac{\mathbb{E}_i\, D_{\mathrm{NN}}(\hat{z}^{\mathrm{ol}}_{i,h})}{\mathbb{E}_i\, D_{\mathrm{NN}}(z_{i,h})}. \tag{6}$$

True future test latents provide the baseline distance to $\mathcal{Z}_{\mathrm{ref}}$. Thus, $R_{\mathrm{NN}}(h) > 1$ means predicted latents lie farther from the empirical encoded-latent manifold than true future latents at the same horizon.

Fourth, *transition magnitude* measures how much the open-loop latent changes between consecutive retained frames. We compare the average predicted step size with the average true encoded step size:

$$\rho^{\mathrm{ol}}_{\mathrm{vel}}(h) = \frac{\mathbb{E}_i \left\| \hat{z}^{\mathrm{ol}}_{i,h} - \hat{z}^{\mathrm{ol}}_{i,h-5} \right\|_2}{\mathbb{E}_i \left\| z_{i,h} - z_{i,h-5} \right\|_2}. \tag{7}$$

A ratio of 1 means the rollout moves at the same latent speed as the true encoded trajectory; values below 1 indicate that the rollout updates too slowly.

Finally, *directional persistence* is the mean cosine similarity between consecutive latent steps, computed for both predicted and true encoded trajectories. Larger values indicate smoother, more directionally persistent motion. LeWM tracks this temporal-straightness quantity during training [7]; we measure it during long-horizon rollout.

C. Teacher-forced control

Teacher forcing separates local one-step error from open-loop compounding by replacing predicted context latents with real encoded latents [21]. We evaluate teacher-forced and open-loop predictions on the same held-out episodes, recorded expert actions, and horizons. In open-loop rollout, the predictor reuses its own previous predictions as context. In teacher forcing, every context latent is a real encoded latent. Thus, any deficit that remains under teacher forcing cannot be attributed to feeding predicted latents back into the model. We measure the teacher-forced transition magnitude ratio:

$$\rho^{\mathrm{tf}}_{\mathrm{vel}}(h) = \frac{\mathbb{E}_i \left\| \hat{z}^{\mathrm{tf}}_{i,h} - z_{i,h-5} \right\|_2}{\mathbb{E}_i \left\| z_{i,h} - z_{i,h-5} \right\|_2}. \tag{8}$$

A value near 1 means the one-step teacher-forced prediction moves at the same latent speed as the true encoded transition;

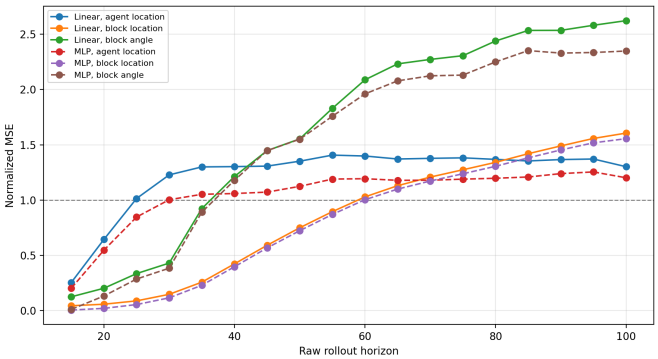


Fig. 1. Probe NMSE (Eq. 5) on open-loop rollout latents for the three components of s_t, using MLP and linear probes. NMSE $= 1$ marks the constant-mean baseline. Agent location crosses the baseline earliest, block location degrades more gradually, and block angle shows the sharpest transition.

values below 1 mean the predicted step is too short. If long-horizon degradation were mainly caused by open-loop compounding, this ratio should be close to 1 under teacher forcing. If it remains low, the transition under-updating is already present in the one-step prediction.

We further test whether the small teacher-forced transition magnitude is simply a shorter step in the correct direction. For each episode and horizon, the reverse triangle inequality gives:

$$\left\| \hat{z}^{\mathrm{tf}}_{i,h} - z_{i,h} \right\|_2 \geq \left| \left\| z_{i,h} - z_{i,h-5} \right\|_2 - \left\| \hat{z}^{\mathrm{tf}}_{i,h} - z_{i,h-5} \right\|_2 \right|. \tag{9}$$

The bound is the minimum target error possible for the predicted step length. Equality holds only when the predicted step is collinear with the true step. Error above the bound means the predicted step is not only too short, but also partly misdirected.

V. Findings

All results below are computed on the 1,000 held-out test episodes described in Section IV. MLP probes are the primary physical-state readout; linear probes are used as a simpler consistency check.

A. Three degradation patterns in decoded physical state

Figure 1 applies the calibrated probes from Table I to predicted open-loop latents. Error rises with horizon for all three physical quantities, but the three curves do not fail in the same way. Agent location crosses the constant-mean baseline by h=30 and then changes little, staying between 1.05 and 1.25 for the rest of the rollout. Block location degrades more gradually: it remains below 0.23 through h=35, crosses the baseline near h=60, and reaches 1.56 at h=100. Block angle has the sharpest transition. It remains below the baseline through h=30 (NMSE $\leq$ 0.38), rises to the threshold region by h=35, and reaches about 2.35 by the final horizon. Linear probes show the same ordering, with earlier baseline crossings.

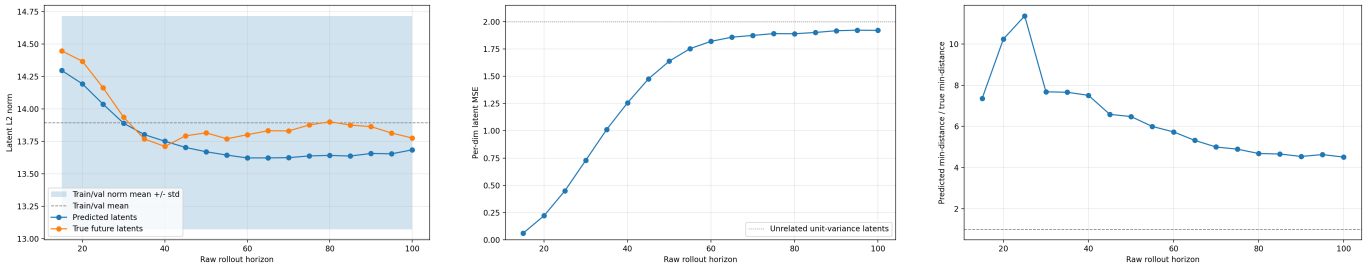


Fig. 2. Stable scale does not imply target fidelity. (a) Predicted latent norms remain inside the encoded reference band. (b) Per-dimension error to the true future latent rises to 1.92. (c) Predicted latents are farther from the empirical encoded reference set than true future latents at the same horizons.

B. Latents remain well-scaled but lose target alignment

By latent norm, the rollouts look healthy. Predicted norms stay between 13.62 and 14.30 at every horizon, against reference statistics of $\mu_{\text{ref}} = 13.89$ and $\sigma_{\text{ref}} = 0.82$; the z-score never leaves $[-0.33, +0.49]$ (Figure 2a). On scale alone, a horizon-100 prediction is indistinguishable from an encoder output. This rules out gross norm collapse or explosion. It does not show that the predicted latent corresponds to the right future state.

Target alignment shows the mismatch directly. The per-dimension squared distance to the true future encoded latent grows from 0.062 at h=15 to 1.92 at h=100 (Figure 2b). Because the encoded dimensions are close to unit variance, this late-horizon value is on the scale expected for two unrelated unit-variance latent vectors. The predicted latent is therefore not just slightly displaced from the correct future; by late rollout, it has lost most episode-specific target alignment.

The nearest-neighbour diagnostic adds the distributional part of the picture. Across horizons, predicted latents are about 4.5 to 11 times farther from the empirical train/validation encoded reference set than true future test latents at the same horizon (Eq. 6) (Figure 2c). The ratio is not monotone: it peaks near 11 around h=25 and decreases to about 4.5 by h=100. Because $R_{\text{NN}}(h)$ is normalised by the true-latent nearest-neighbour distance, this decrease is consistent with later true futures being less densely covered by the finite reference set, rather than with recovery of the predicted rollout.

In absolute terms, though, the predicted latents remain close to the encoded distribution. At h=100 the nearest encoded latent is about 0.09 per dimension away, while the correct future is 1.92 away. The rollout therefore sits within plausible regions of latent space while losing the particular future it should represent.

C. Predicted trajectories move too slowly and too straight

The previous diagnostics compare where the rollout lands. The trajectory diagnostics show how it moves. Predicted rollouts update slowly and change direction less than true encoded trajectories.

Open-loop predicted steps are much shorter than the corresponding encoded steps. True encoded steps average about 4.0 latent units at early horizons and grow to about 5.8 as the trajectory becomes more active. Predicted open-loop steps start near 1.1 and drift toward 0.9 (Figure 3a), so $\rho_{\text{vel}}^{\text{ol}}$ (Eq. 7) falls from 0.32 to 0.16.

Predicted trajectories are also straighter than true encoded trajectories at every horizon (Figure 3b). The mean cosine between consecutive steps is 0.70 for predicted rollouts, compared with 0.53 for true encoded trajectories. Thus the predicted rollout is not only slower; its direction also varies less over time.

A trajectory that is too short and too straight admits two interpretations. The one-step predictor may damp motion even when given real encoded inputs, or the damping may arise only after the predictor starts consuming its own imperfect outputs. Open-loop measurements cannot separate these effects because both are present at once. The teacher-forced control does separate them.

D. The transition under-updating is intrinsic to the one-step prediction

The teacher-forced control separates local one-step prediction error from errors induced by autoregressive feedback. If the transition under-updating were mainly caused by feeding predictions back into the context, then replacing those predictions with real encoded latents should restore the predicted/true transition magnitude ratio toward 1. It does not. The teacher-forced ratio in Eq. 8 stays between 0.317 and 0.339 across all eighteen horizons (Figure 4a). Even when the predictor is handed exact encoded context at every step, its predicted step is only about one-third as long as the true encoded transition.

Open-loop feedback further reduces transition magnitude over the rollout. The open-loop transition magnitude ratio starts at the same level as the teacher-forced ratio at the first predicted step, then falls to about 0.16 by late rollout. At h=100, the teacher-forced transition magnitude shortfall is approximately $1 - 0.33 = 0.67$, while the open-loop shortfall is approximately $1 - 0.16 = 0.84$; the local one-step deficit therefore accounts for about $0.67/0.84 \approx 0.80$ of the final transition magnitude shortfall. This attribution is heuristic because teacher-forced and open-loop shortfalls are measured under different context distributions. The value refers only to transition magnitude shortfall, not to total target error.

The target-error split confirms that distinction (Figure 4b). Under teacher forcing, the per-dimension one-step error grows only from about 0.062 to 0.139 across horizons. A copy-last

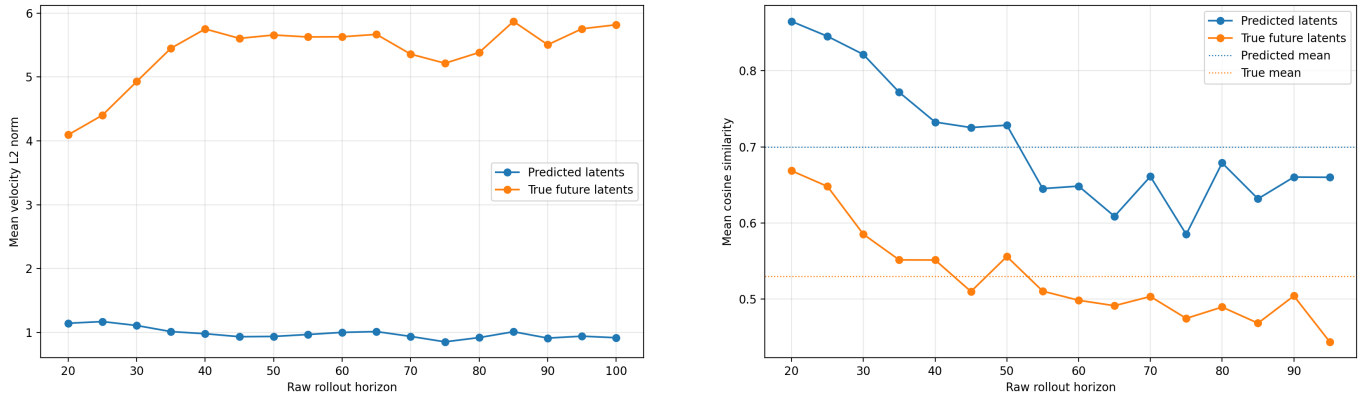


Fig. 3. Open-loop trajectories are under-updated and over-smoothed. (a) Predicted latent steps are several times shorter than true encoded steps, with the transition magnitude ratio falling from about 0.32 to about 0.16. (b) Predicted trajectories are consistently straighter than true encoded trajectories. Both panels are plotted from h=20, since these trajectory metrics are computed from increments between consecutive retained latent states rather than from a single rollout state.

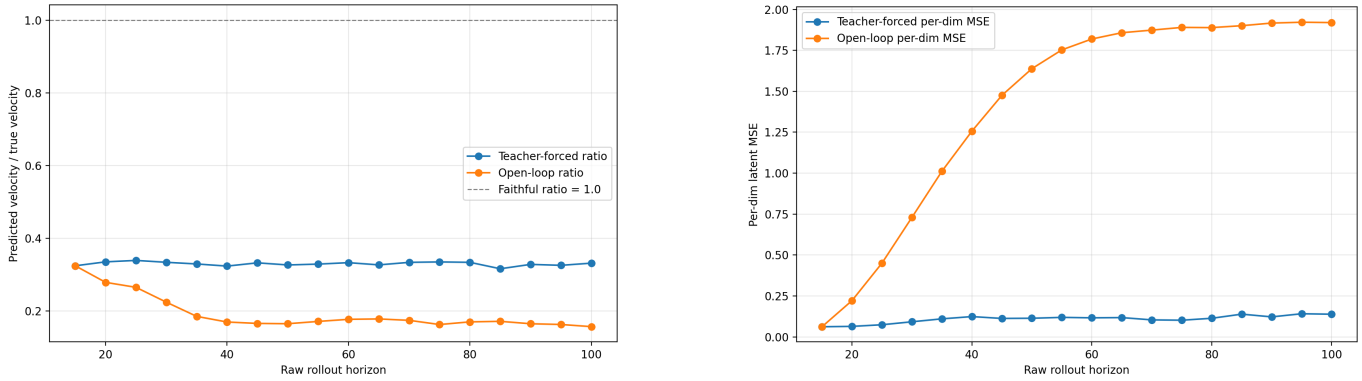


Fig. 4. Teacher forcing separates local under-updating from open-loop feedback. (a) The teacher-forced transition magnitude ratio stays near one-third across all horizons, while the open-loop ratio decays toward one-sixth. (b) Teacher-forced one-step error remains small compared with open-loop target error, showing that feedback governs final displacement while local transition-magnitude under-updating is already present under exact context.

baseline that predicts $\hat{z}_t := z_{t-5}$ gives per-dimension error between 0.094 and 0.225 across horizons. The teacher-forced predictor reduces this to 0.60 to 0.66 of the copy-last error, so the one-step prediction is not a near-identity shortcut. Under open-loop rollout, the per-dimension error reaches 1.92 by h=100. Open-loop compounding therefore largely determines the final distance from the correct future latent, while teacher forcing shows that the local step is already under-updated before compounding begins.

The short step is not simply a shorter version of the correct step. Equation 9 gives the smallest target error possible for the observed predicted step length if the predicted step were collinear with the true step. The measured teacher-forced error exceeds this lower bound at every horizon. At h=55, for example, the true step averages about 5.63 latent units and the teacher-forced predicted step about 1.85, giving a pure-length lower bound of about 3.78. The measured one-step error is about 4.32. The excess, roughly 0.5 to 0.65 latent units across horizons, implies that the predicted step is partly misdirected as well as too short.

Together, the diagnostics identify a specific failure mode. The rollout does not primarily fail by exploding, collapsing, or leaving latent space outright. It stays globally well-scaled and broadly latent-like, but it loses alignment with the episode-specific future. Its trajectory is under-updated, straighter than the true encoded trajectory, and locally damped even under teacher forcing. These results localise the damping to the one-step predictor; possible causes and limitations are discussed next.

VI. Discussion and Limitations

a) Interpretation: The findings point to a local under-updating effect rather than a simple off-manifold failure. One plausible interpretation is conditional-mean smoothing. A deterministic predictor trained with a squared latent prediction loss can reduce error by averaging over plausible next latents when the future is ambiguous. In pixel-space video prediction, the analogous effect is known to produce blurry forecasts under MSE-style objectives [11]. In latent space, we find the corresponding pattern directly: predicted rollouts take smaller steps, follow straighter paths, and remain broadly plausible

while missing the correct future. These diagnostics support the smoothing interpretation at the level of rollout geometry, but they do not by themselves identify which design choice produces it. Future work could use controlled ablations to examine how the loss function, model architecture, action conditioning, and data ambiguity contribute to transition damping, and whether reducing this effect improves rollout fidelity or downstream planning.

b) Relation to LeWM's operating regime: These diagnostics should not be read as contradicting LeWM's reported planning success. LeWM plans over five predictor steps, ending at $h{=}35$ under our indexing, and within this range the block-state probes remain largely informative, although agent location and latent transition magnitude already show degradation. Replanning can mask some open-loop error by re-encoding real observations, and a planner may still rank candidate action sequences correctly under a roughly systematic shrinkage of latent motion. We therefore interpret the longer horizons as a stress test of rollout fidelity beyond the demonstrated operating regime, relevant to longer plans, fewer replans, and settings where predicted latent rollout segments must remain accurate without frequent re-encoding from real observations.

c) Limitations: The study is intentionally scoped as a controlled case study of one released checkpoint and one task. Push-T is useful because its compact physical state and contact-rich dynamics make rollout degradation measurable. The results give direct evidence about LeWM on Push-T; whether the same pattern generalises is an empirical question for the same diagnostics across other checkpoints, tasks, and model families. The probe results are also readout-specific: they measure whether predicted latents preserve state decodability under probes calibrated on real encoded latents, rather than all information that may be present in the latent representation. Because predicted latents gradually move away from the distribution seen during probe training, declining probe accuracy may partly reflect this distribution shift rather than necessarily indicating that physical information has been lost from the latent representation.

VII. Conclusion

We introduced a rollout-level diagnostic framework and applied it to LeWM on Push-T. Predicted latents remain well-scaled while losing target alignment, physical-state decodability, and faithful trajectory geometry. Teacher forcing shows that transition-magnitude under-updating is already present locally, while recursive rollout adds substantial target error. These results motivate direct evaluation of predicted latent trajectories alongside representation and planning metrics.

References

[1] D. Ha and J. Schmidhuber, "World models," *CoRR*, vol. abs/1803.10122, 2018.

[2] D. Hafner *et al.*, "Learning latent dynamics for planning from pixels," in *ICML*, ser. Proceedings of Machine Learning Research, vol. 97. PMLR, 2019, pp. 2555–2565.

[3] D. Hafner, T. P. Lillicrap, J. Ba, and M. Norouzi, "Dream to control: Learning behaviors by latent imagination," *CoRR*, vol. abs/1912.01603, 2019.

[4] M. Assran *et al.*, "Self-supervised learning from images with a joint-embedding predictive architecture," in *CVPR*. IEEE, 2023, pp. 15 619–15 629.

[5] A. Bardes *et al.*, "Revisiting feature prediction for learning visual representations from video," *Trans. Mach. Learn. Res.*, 2024.

[6] M. Assran *et al.*, "V-JEPA 2: Self-supervised video models enable understanding, prediction and planning," 2025, arXiv preprint arXiv:2506.09985.

[7] L. Maes, Q. L. Lidec, D. Scieur, Y. LeCun, and R. Balestriero, "LeWorldModel: Stable end-to-end joint-embedding predictive architecture from pixels," 2026, arXiv preprint arXiv:2603.19312.

[8] G. Zhou, H. Pan, Y. LeCun, and L. Pinto, "DINO-WM: world models on pre-trained visual features enable zero-shot planning," in *ICML*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR / OpenReview.net, 2025.

[9] M. Janner, J. Fu, M. Zhang, and S. Levine, "When to trust your model: Model-based policy optimization," *CoRR*, vol. abs/1906.08253, 2019.

[10] N. Lambert, K. Pister, and R. Calandra, "Investigating compounding prediction errors in learned dynamics models," 2022, arXiv preprint arXiv:2203.09637.

[11] M. Mathieu, C. Couprie, and Y. LeCun, "Deep multi-scale video prediction beyond mean square error," in *International Conference on Learning Representations*, 2016.

[12] V. Sobal, W. Zhang, K. Cho, R. Balestriero, T. G. J. Rudner, and Y. LeCun, "Learning from reward-free offline data: A case for planning with latent dynamics models," 2025, arXiv preprint arXiv:2502.14819.

[13] K. Zhao *et al.*, "Sub-JEPA: Subspace gaussian regularization for stable end-to-end world models," 2026, arXiv preprint arXiv:2605.09241.

[14] W. Zhang *et al.*, "Hierarchical planning with latent world models," 2026, arXiv preprint arXiv:2604.03208.

[15] Z. Duan *et al.*, "LiveWorld: Simulating out-of-sight dynamics in generative video world models," 2026, arXiv preprint arXiv:2603.07145.

[16] G. Alain and Y. Bengio, "Understanding intermediate layers using linear classifier probes," 2018, arXiv preprint arXiv:1610.01644.

[17] J. Hewitt and P. Liang, "Designing and interpreting probes with control tasks," 2019, arXiv preprint arXiv:1909.03368.

[18] R. Upadhyay *et al.*, "WorldBench: Disambiguating physics for diagnostic evaluation of world models," 2026, arXiv preprint arXiv:2601.21282.

[19] R. Balestriero and Y. LeCun, "LeJEPA: Provable and scalable self-supervised learning without the heuristics," 2025, arXiv preprint arXiv:2511.08544.

[20] C. Chi *et al.*, "Diffusion Policy: Visuomotor policy learning via action diffusion," 2024, arXiv preprint arXiv:2303.04137.

[21] R. J. Williams and D. Zipser, "A learning algorithm for continually running fully recurrent neural networks," *Neural Comput.*, vol. 1, no. 2, pp. 270–280, 1989.